

Spatiotemporal analysis of PM_{2.5} in Nanjing, China, using network theory and PCA

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Abstract: This study investigates the spatial and temporal correlation patterns of PM_{2.5} concentrations recorded by over 280 air quality monitoring stations in Nanjing, China, over the course of one month (December 2019). Employing a combination of Spearman correlation analysis, network analysis, and Principal Component Analysis (PCA), we explore the dynamic connectivity between stations over the diurnal cycle. The objective is to determine whether the city behaves as a spatially cohesive unit or as a collection of localized communities in terms of PM_{2.5} levels. The analysis identifies critical hours with maximum correlation coherence and reveals periods of spatial fragmentation. This approach provides insights that can inform data-driven environmental management and urban air quality assessment.

Keywords: Air quality, Spearman correlation, Network analysis, PM_{2.5}, PCA, Environmental informatics

1 Introduction

Air pollution, particularly due to fine particulate matter (PM_{2.5}) is a critical concern in urban areas worldwide, with documented impacts on human health, visibility and climate. PM_{2.5} particles, which have a diameter of less than 2.5 micrometres, can penetrate deep into the respiratory tract and even enter the bloodstream, causing cardiovascular diseases. As urbanization accelerates and vehicular emissions increase, the need for detailed, high-resolution analysis of pollution patterns has become paramount. The ability to quantify and visualize how pollution behaves over time and space is essential for city-wide planning and environmental health strategies.

Traditionally, air quality monitoring has relied on statistical summaries, time-series decomposition, or interpolation techniques. While valuable, these approaches often fail to capture the dynamic and interconnected nature of pollution across dense urban networks. Emerging methodologies, inspired by data science and network theory provide a more nuanced framework. Spearman correlation, for example, allows for non-linear relationships detection between sensor outputs, and Network Analysis (NA) can reveal

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hidden structural patterns in monitoring data, such as community structures or influential hubs [DZ17]. Principal Component analysis (PCA), on the other hand, aids in identifying latent temporal patterns and reducing the complexity of high-dimensional datasets.

In this study, we integrate these approaches to examine an extensive PM_{2.5} monitoring dataset from Nanjing, China. Our aim is to assess whether the city's air quality functions as a cohesive single network or as distinct localized sub-networks, and to pinpoint key time periods that exhibit either maximum coherence or significant fragmentation in pollution patterns.

2 Materials and methods

2.1 Study area and data sources

The analysis is centred on Nanjing, a major city in eastern China. The dataset contains hourly concentrations of 7 pollutants (NO₂, SO₂, CO, O₃, TVOC, PM₁₀, PM_{2.5}) over a full calendar month (December of 2019), from 282 air quality monitoring stations across the city, while no additional data (like meteorological observations) were made available. The Cambri AER2 - series (Shenzhen Cambri Environmental Technology Ltd., China) is the air quality monitoring device used. The dense and widespread spatial distribution of these devices (Figure 1) makes Nanjing an ideal case for high-resolution correlation and network analysis. This means that we address these devices as a network where each unit represents a node (that has attributes, i.e. concentration values of pollutants of interest), and the lines that may be envisaged connecting all those nodes are the network edges (representing potential relationships or connections between the devices).

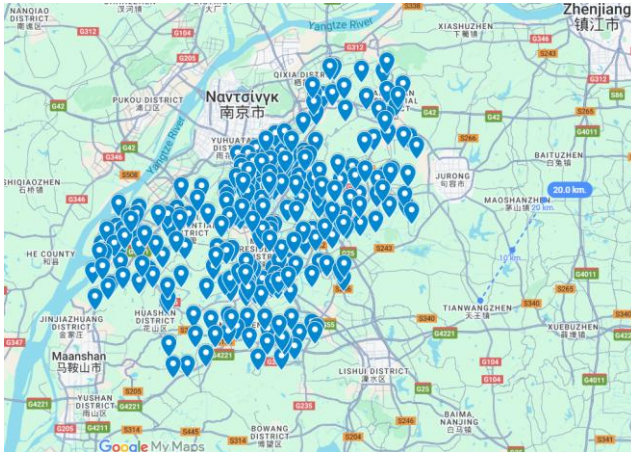


Fig. 1: Geospatial Distribution of the Air Quality Monitoring Stations Across Nanjing. An indicative scale has been included for distance estimation.

2.2 Methodology

Data preprocessing

Data cleaning involved filtering out rows with missing or non-relevant pollutant values. Timestamp fields were converted to Python datetime objects and decomposed into separate 'Date', 'Hour', and 'DateHour' columns. Concentration fields were cast to numeric types to ensure proper aggregation and correlation calculations.

Metrics and plots

We computed descriptive statistics (mean, standard deviation, minimum, maximum, skewness, and kurtosis) of concentrations across all stations for each pollutant and for each hour of each day. Boxplots were generated for each pollutant and date to visualize diurnal patterns and variability. In addition, and to evaluate the similarity in pollutant behaviour across stations, Spearman rank correlation matrices were computed for each hour of the day. Each matrix represents pairwise correlations of PM_{2.5} values between all stations at that specific hour, reflecting how synchronized or divergent the pollutant dynamics are across the network. For each hour, the mean Spearman correlation of each station with all others was also calculated. This yielded a 282×24 matrix that characterises the temporal “centrality” or influence of each station. Stations with higher average correlations are those that consistently align with broader pollution trends. We then identified the top 10 stations with the highest average correlation at each hour therefore providing with a temporal view of the most synchronized stations in the pollution network.

Next, we examined the off-diagonal entries of the hourly correlation matrices to derive a 24-hour profile of spatial coherence. Excluding the main diagonal (self-correlations) and any perfect correlations ($|r| = 1$), we computed the average inter-station correlation for each hour [ZM21]. This temporal coherence highlights the hours of highest and lowest spatial synchronization in PM_{2.5} levels across the city.

Principal Component Analysis

For each hourly correlation matrix, the upper triangular off-diagonal values were extracted and flattened into a vector representing that hour’s spatial correlation profile. We applied a threshold of $r > 0.4$ to each vector to filter out weak correlations. The resulting vectors from all 24 hours were then combined into a $24 \times N$ matrix, where N is the total number of station pairs with $r > 0.4$ in at least one hour. We performed PCA on this matrix, retaining 10 PCs to capture the dominant modes of variability [Kr23]. A scree plot was used to examine the variance explained by each component, and a 3D scatter plot of the first three principal components was generated to visualize the clustering of hours and shifts in station behaviour in the reduced-dimensional space.

Network Analysis

For each hour, we constructed a network graph where nodes represent monitoring stations and edges connect station pairs with Spearman correlation $\rho > 0.6$ (using the correlation

value as the edge weight). In environmental data analysis, Spearman correlation values above 0.6 are generally considered to indicate moderate to strong monotonic associations. Such filtering has precedent in atmospheric and sensor network studies, where correlations thresholds between 0.5 and 0.7 are commonly used [LI22].

Key network metrics (node degree, density etc) are computed to assess the overall connectivity and complexity of the system. Community detection identifies densely connected groups of stations (subnetworks) that behave similarly in terms of PM_{2.5} dynamics. These graph-theoretic metrics quantify the evolving structural behaviour of the station network over time [VI21].

3 Results and discussion

As part of the initial statistical exploration, descriptive metrics for each pollutant were calculated on an hourly basis. Fig. 2 shows a snapshot of the computed values for PM_{2.5}. Each row corresponds to a specific hour and includes the mean, standard deviation, minimum, maximum concentrations, along with skewness and kurtosis. These indicators offer a multifaceted view of the pollutant’s temporal dynamics. The data reveal a clear diurnal pattern. The specific PM_{2.5} levels are generally higher during early morning (e.g., 0:00–6:00) and late afternoon to evening hours (16:00–23:00), which correlates with increased human activity, traffic congestion, and possibly industrial operations. For instance, at 18:00 on Dec 1st, the mean concentration peaks at 68.7 $\mu\text{g}/\text{m}^3$, indicating a potential pollution episode.

Pollutant	Date	Hour	Mean	Std Dev	Min	Max	Skewness	Kurtosis
PM _{2.5}	2019-12-01	0	39,74206	8,846666	20,37373	66,89312	0,73731	0,255115
		1	35,47724	7,996032	18,12749	63,4289	0,62517	-0,12525
		2	35,65992	6,345172	21,96897	61,15474	0,86708	0,960845
		3	33,49925	8,058019	20,47161	61,3834	1,18887	1,293207
		4	30,85923	7,229466	18,67006	62,49007	1,379422	2,660828
		5	28,78656	8,186775	14,42691	54,60121	0,518282	-0,10337
		6	26,82668	7,531809	15,07086	47,09575	0,566218	-0,68448
		7	23,96541	6,593224	14,51925	51,80502	1,651838	3,15972
		8	27,04202	6,667591	15,00445	48,83425	1,021028	1,022579
		9	27,52143	6,332253	14,12711	67,94593	1,602616	6,285344
		10	27,94723	5,213968	14,91794	67,76965	1,707589	11,47429
		11	26,76644	4,980806	15,3346	54,53233	0,925032	3,217375
		12	26,27236	4,233085	17,06871	55,31897	1,512813	7,992318
		13	27,21015	4,166982	17,26887	50,1081	0,986952	4,321658
		14	29,45257	4,416918	17,37753	43,75729	0,273278	0,609428
		15	30,14754	4,152865	18,27735	42,79836	0,040964	0,284829
		16	55,63405	9,209799	31,05306	103,1533	0,760544	2,381652
		17	40,04287	3,745617	29,18513	59,4248	0,250166	2,391453
		18	68,73554	12,59546	36,2603	126,8273	0,257479	1,604599
		19	32,61269	3,972184	21,39698	50,09524	0,030633	2,350793
		20	38,66311	5,570438	23,02882	68,68296	0,517526	3,938416
		21	42,02836	5,611769	23,51782	70,20358	0,011411	3,928548
		22	44,92726	6,058244	25,24972	82,63899	0,413139	6,467839
		23	46,30589	6,631792	27,48297	91,38252	1,120877	8,573812

Fig. 2: Hourly Descriptive Statistics for PM_{2.5} concentrations on December 1–2, 2019

Figure 3 illustrates the diurnal variation of PM_{2.5} concentrations for a representative day (December 7, 2019), shown as boxplots of the hourly concentration distribution across all stations. Each boxplot summarizes the central tendency, dispersion and outliers for a given hour of the day. The results highlight a clear diurnal pattern:

- Morning to midday (7:00-12:00), concentrations begin to increase steadily, peaking around 9:00 to 10:00. This aligns with increased traffic and reduced atmospheric dispersion.
- Afternoon (13:00-16:00), a moderate decline is observed, possibly due to improved dispersion conditions.
- Evening (18:00-23:00), a second rise in concentrations is evident, particularly after 19:00, reflecting the evening rush hour and accumulation effects under cooling atmospheric conditions.

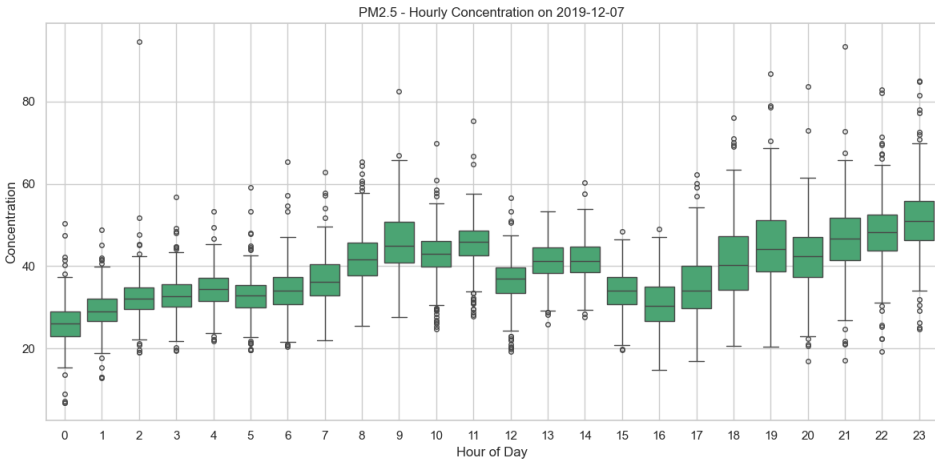


Fig. 3: Diurnal Variation of PM_{2.5} concentrations across stations in Nanjing, on December 7, 2019.

The Spearman correlation matrix that was calculated provided with valuable insights. Figure 4 presents the diurnal evolution of the average Spearman correlation coefficient among all monitoring stations in Nanjing for PM_{2.5} concentrations. The plot excludes perfect correlations (± 1) and captures only meaningful off-diagonal relationships. This indicator reflects the overall degree of spatial coherence - i.e., how similarly pollutant levels fluctuate across the city during each hour of the day.

Key observations include:

- High coherence throughout the day: From 01:00 to 16:00, the average ρ remains consistently above 0.85, peaking between 10:00 and 13:00. This suggests that during daylight hours, stations exhibit strong synchrony in pollutant patterns—likely due to shared exposure to emissions, meteorological uniformity, or traffic flows.

- Sharp decline in the evening: After around 19:00, a marked decrease in interstation correlation is observed, reaching a minimum around 21:00. This drop may reflect the influence of localized sources (e.g., residential heating), boundary layer breakup, or other nocturnal atmospheric effects leading to heterogeneous $\text{PM}_{2.5}$ behaviour.
- Late-night recovery: The average correlation begins to rebound after midnight, indicating a partial restoration of uniform pollutant dynamics.

These findings highlight the non-stationary structure of spatial correlation over time and provide motivation for hour-by-hour network modelling. The evening period emerges as a phase of spatial fragmentation, warranting further analysis of network modularity and community formation.

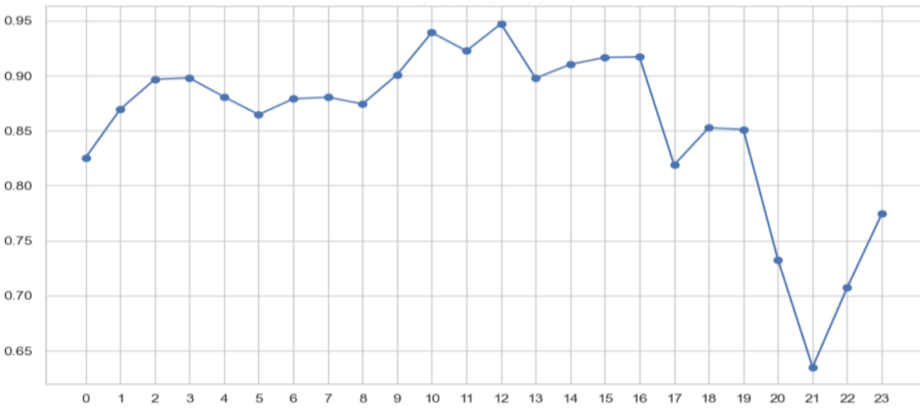


Fig. 4: Hourly profile of mean Spearman Correlation for $\text{PM}_{2.5}$ among all monitoring stations in Nanjing. Horizontal axis depicts the hour of the day (in local time) while the vertical axis presents with the correlation coefficient value.

The analysis continued with the calculation of the PCA based on the Spearman correlation matrices as presented in the prev. chapter. For each hour, the upper triangle of the correlation matrix—comprising all non-redundant, off-diagonal values—was extracted and flattened into a vector. These vectors represent the spatial configuration of $\text{PM}_{2.5}$ coherence at each hour and were used to construct a $24 \times N$ matrix, where N is the number of unique station-to-station relationships after thresholding. The scree plot in Figure 5 shows the proportion of explained variance for the first 10 principal components. It is evident that PC1 to PC3 account for approximately 60–65% of the total variance, indicating that a significant portion of the temporal variability in spatial correlation structure can be captured using just three dimensions. For this reason, and in Figure 6, a 3D scatter plot visualizes each hour of the day as a point in the PC1–PC3 space. On this basis hours 0–19 appear to be relatively clustered, indicating that during most of the day the spatial structure of $\text{PM}_{2.5}$ correlations among all monitoring stations remains similar. In contrast, hours 20–23 diverge significantly from the main cluster, implying a distinct spatial network configuration in the late evening. These hours correspond to periods

previously identified as having lower average interstation correlations and higher modularity. The separation of these late hours in PC space supports the hypothesis of a temporal shift in the network’s coherence, possibly due to local emission sources, boundary layer changes, or reduced traffic homogenization effects.

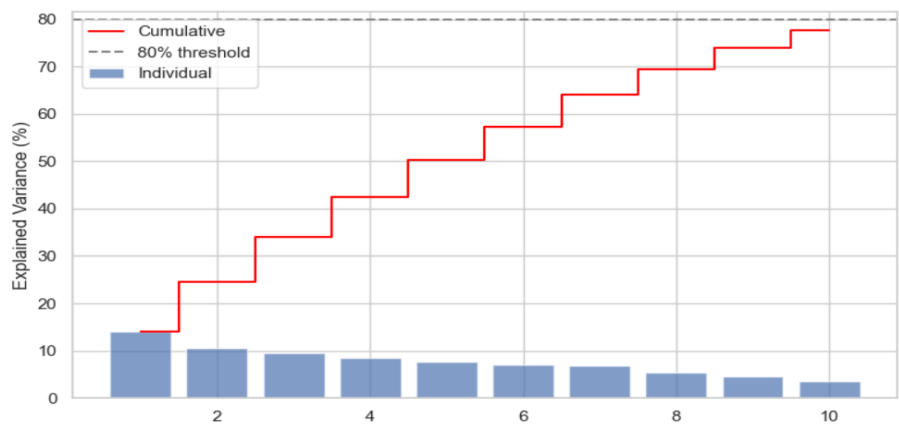


Fig. 5: Scree Plots Showing explained variance of the first 10 PCA components.

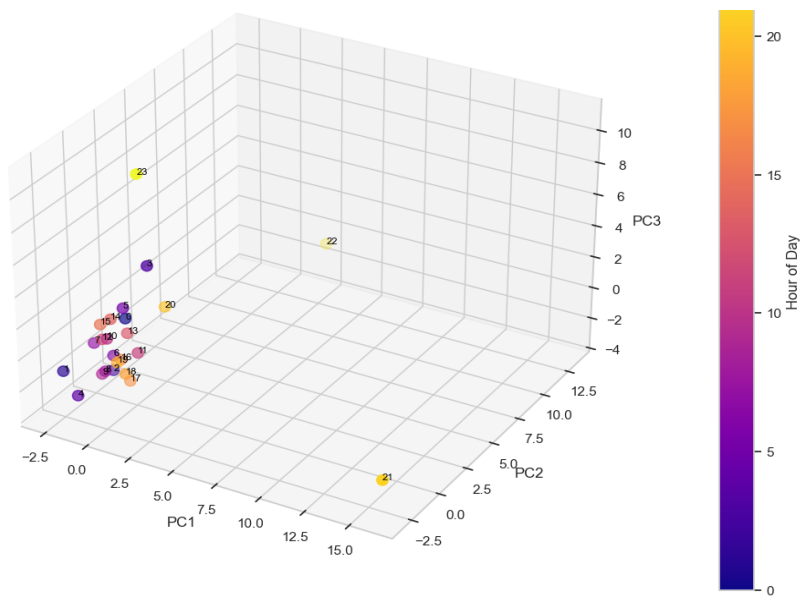


Fig. 6: 3D Visualization of PCA components (PC1, PC2, PC3)

Finally, we examined the graph-theoretic properties of the PM_{2.5} correlation network at each hour (using $\rho > 0.6$ to define edges). Key network metrics are summarized as follows:

- **Node count:** The number of active nodes (stations) remains nearly constant at about 281 throughout the day, indicating that almost all stations reported valid data at all hours.
- **Edge count and average degree:** Both metrics reach their highest values during the late morning and early afternoon (approximately 10:00–15:00), reflecting a very well-connected network of stations at those times and thus high overall spatial coherence,
- **Network density:** Network density exceeds 0.99 during several midday hours, meaning nearly every possible station pair is connected by a strong correlation. This confirms the uniformly coherent behaviour of the pollution network in the middle of the day.
- **Community count:** For most hours, the correlation network splits into only 2–3 communities, indicating minimal fragmentation of the city into independent sub-regions. However, in the late evening (around 21:00–23:00) the number of communities rises (up to 4), signifying a resurgence of fragmentation as the station network divides into more distinct clusters.
- **Modularity:** Consistent with the above, modularity values are extremely low (< 0.02) during the high-coherence midday period, which is typical of a strongly interconnected network with little community structure. In the evening, by contrast, modularity increases sharply (for example, reaching ~ 0.23 at 22:00), suggesting a phase of spatial disintegration where stations form well-defined but weakly interconnected subgroups [Fe20].

Figure 7 presents a comprehensive table of graph-theoretical metrics computed from hourly Spearman correlation networks for PM_{2.5}, using a correlation threshold of $\rho > 0.6$ to define edges. Each row represents one hour of the day, and the values are derived from the corresponding undirected weighted graph of interstation connections.

- Edge count and average degree are highest during midday (e.g., hours 10–15), suggesting a well-connected sensor network with strong overall correlations.
- Network density exceeds 0.99 in several hours, especially between 10:00 and 15:00, further confirming uniform correlation behaviour during this interval.
- Community count (NumCommunities) remains low (2–3) for most hours, indicating limited fragmentation in the sensor network. However, late evening hours (e.g., 21–23) see a resurgence of fragmentation with higher community counts (up to 4).
- Modularity values remain low (< 0.02) during high-coherence periods, which is typical for strongly interconnected networks. In contrast, sharp increases in modularity appear during the evening and early night (e.g., hour 22: modularity = 0.23), suggesting a phase of spatial disintegration.

Hour	Nodes	Edges	AvgDegree	Density	NumCommunities	Modularity
0	281	37390	266,1209964	0,950432	3	0,040478585
1	281	38566	274,4911032	0,980325	3	0,021599129
2	281	38272	272,3985765	0,972852	4	0,00771696
3	281	38379	273,1601423	0,975572	4	0,011079856
4	281	37594	267,5729537	0,955618	3	0,014089596
5	280	37486	267,7571429	0,959703	3	0,014201657
6	280	38279	273,4214286	0,980005	2	0,011679466
7	280	38260	273,2857143	0,979519	2	0,012698059
8	280	37998	271,4142857	0,972811	2	0,017238184
9	280	38575	275,5357143	0,987583	2	0,008853348
10	281	39301	279,7224199	0,999009	3	0,005135129
11	281	39206	279,0462633	0,996594	2	0,012423838
12	281	39217	279,1245552	0,996873	2	0,007916528
13	281	39143	278,5978648	0,994992	2	0,015234877
14	281	39302	279,7295374	0,999034	3	0,012561559
15	281	39311	279,7935943	0,999263	2	0,008033067
16	281	38910	276,9395018	0,98907	2	0,009260324
17	281	36535	260,0355872	0,928699	2	0,039320643
18	281	38558	274,4341637	0,980122	3	0,020669806
19	281	38176	271,7153025	0,970412	2	0,022721893
20	281	31404	223,5160142	0,798271	2	0,071487052
21	281	22884	162,8754448	0,581698	3	0,232371444
22	281	28554	203,2313167	0,725826	4	0,098759449
23	281	34135	242,9537367	0,867692	3	0,055148293

Fig. 7: Hourly evolution of network metrics based on PM2.5 correlation

4 Conclusions

This study demonstrates the utility of combining correlation analysis, PCA, and network analysis to gain insights into the spatiotemporal structure of urban PM_{2.5} pollution. Using hourly Spearman correlation matrices (and their upper-triangular values), we characterized the dynamic inter-station relationships of PM_{2.5} throughout the day in Nanjing, China.

The results indicate a cyclic behaviour of the sensor network's coherence: midday hours revealed strong uniformity with dense graph structures and low modularity, while late evening hours marked significant fragmentation, both in terms of lower mean correlations and rising community separation. PCA provided dimensionality reduction that confirmed these structural shifts, particularly isolating hours 20–23 as outliers in the system's temporal evolution. This aligns with possible real-world causes such as shifts in emission sources, traffic decay, or boundary layer transitions.

Overall, our analytical approach offers a replicable and scalable framework for treating an urban air quality monitoring network not just as a passive collection of sensors, but as a dynamic system with distinct temporal modes of behaviour. Such insights can support

targeted interventions, adaptive monitoring strategies, and early-warning systems for pollution clustering events.

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